

Nature Code/Software Submission Checklist — FILLED

“Same Prompt, Different Answer: Hidden Non-Determinism in LLM APIs Undermines Scientific Reproducibility”

Reference policy: Nature code & software submission policy (<https://www.nature.com/documents/nr-software-policy.pdf>). Corresponding author: Lucas Rover (ORCID: 0000-0001-6641-9224) Coauthors: Hugo Valadares Siqueira (0000-0002-1278-4602); Eduardo Tadeu Bacalhau (0000-0002-3936-0375); Anibal Tavares de Azevedo (0000-0003-1678-7795); Yara de Souza Tadano (0000-0002-3975-3419) Manuscript ID: NCOMMS-26-021731A (Major Revision, Nature Communications) Revision date: 2026-05-11

1. Software identification

Item	Response
Software name	GenAI Reproducibility Protocol (Run Card / Prompt Card / W3C PROV reference implementation)
Public repository	https://github.com/Roverlucas/genai-reproducibility-protocol
Repository visibility (review)	Private — reviewer link supplied via Editorial Manager
Repository visibility (post-publication)	Public on acceptance
Versioned release for this manuscript	Tag v1.1 (created at submission of revised manuscript)
Persistent archive (snapshot)	Figshare DOI 10.6084/m9.figshare.31653373 (CC-BY 4.0); GitHub release tag v1.1 is mirrored on Figshare
Reviewer data archive (private)	Figshare reviewer link — see Data Availability for the URL provided to the editorial office (excluded from the published version)
License (code)	MIT (LICENSE file at repository root)
License (data and manuscript artefacts)	CC-BY 4.0
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2. Documentation

Item	Status	Location
README at repository root		README.md (overview, setup, reproducing experiments, citation)
Revision plan and decision log		REVISION_PLAN.md (consolidated response strategy for the Major Revision)
Live status / progress tracker		STATUS.md (per-task status of T1–T17 revision deliverables)
Method-level documentation		Manuscript Methods §Models and infrastructure, §Protocol design, §Statistical analysis; Supplementary §S1–S10
API payload documentation		Supplementary §S4 (exact JSON request bodies for every deployment stack)
Run-Card / Prompt-Card schema		src/protocol/run_card.py, src/protocol/prompt_card.py, JSON schemas under src/protocol/

Item	Status	Location
Provenance documentation (W3C PROV)		<code>src/protocol/prov_generator.py</code> ; PROV-JSON examples in <code>outputs/prov/</code>
Inline code documentation		Module-level and function-level docstrings throughout <code>src/</code>

3. Installation instructions

Item	Response
Programming language and version	Python 3.14.3 (tested); minimum supported 3.10 (declared in <code>README.md</code>)
Package manifest	<code>requirements.txt</code> (pinned versions)
Recommended environment	Virtual environment via <code>python -m venv .venv</code>
Reference install commands	<code>python -m venv .venv && source .venv/bin/activate && pip install -r requirements.txt</code>
Local LLM runtime	Ollama v0.15.5 (https://ollama.com/) — <code>ollama pull llama3:8b mistral:7b gemma2:9b</code>
Optional dependencies	<code>datasets</code> (HumanEval/GSM8K loaders fall back to GitHub-mirror fetch when absent); <code>tiktoken</code> (cost estimator falls back to chars/4 heuristic when absent)
One-command sanity check	<code>python -m pytest tests/ -q</code> (102 tests must pass)

4. Dependencies (from `requirements.txt`)

Package	Pinned version	Purpose
<code>openai</code>	2.17.0	OpenAI API client (GPT-4, GPT-4o stacks)
<code>ollama</code>	0.4.7	Local model serving (LLaMA 3, Mistral, Gemma 2)
<code>numpy</code>	2.4.2	Numerical primitives, bootstrap resampling
<code>python-Levenshtein</code>	0.27.3	Normalized edit distance
<code>pyyaml</code>	6.0.3	Configuration files
<code>tabulate</code>	0.9.0	Tabular reporting in analysis scripts
<code>matplotlib</code>	3.10.8	Figure generation (Fig. 1–4 + Extended Data)
<code>scipy</code>	1.17.0	Fisher’s exact, Mann–Whitney U, Wilcoxon signed-rank
<code>bert-score</code>	0.3.13	BERTScore F1 (semantic-equivalence layer)
<code>datasets</code> (optional)	2.21.0	HumanEval / GSM8K loaders
<code>tiktoken</code> (optional)	0.7.0	Cost estimator

External SDKs called via `urllib` (no extra pip dependency): Anthropic Messages API, Google Gemini API, DeepSeek Chat Completions API, Perplexity Sonar API, Together AI Chat Completions API.

5. Example data and expected output

Item	Path	Notes
AI/ML input corpus (30 abstracts)	<code>data/inputs/abstracts.json</code>	Used for Tasks 1–4
Cross-domain input corpus (10 PubMed PM2.5 abstracts, T14)	<code>data/inputs/revision/</code>	Out-of-AI/ML extension
Run records for Tasks 1–4 (4,104 PROV-instrumented runs)	<code>outputs/runs/</code>	One JSON per run; aggregated in <code>outputs/all_runs.json</code>
Run records for the code/math/cross-domain tasks (2,900 runs; T1+T4+T14)	<code>outputs/revision/runs/</code>	Complete; per-task and per-stack counts reported in manuscript Table 4 and Supplementary §S11
Run Cards (human-readable provenance)	<code>outputs/run_cards/</code> , <code>outputs/revision/run_cards/</code>	One per run
W3C PROV-JSON provenance graphs	<code>outputs/prov/</code>	Validated against W3C PROV-JSON serialization
Aggregated metrics	<code>analysis/expanded_metrics.json</code> , <code>analysis/bootstrap_cis.json</code> , <code>analysis/bertscore_per_field_results.json</code> , <code>analysis/revision/*.json</code>	Inputs for figure/table generators
Expected figure outputs	<code>article/figures/</code>	Compiled PDFs at 600 DPI
Expected table outputs	<code>article/tables/</code> , <code>analysis/tables/</code>	LaTeX fragments included via <code>\input{}</code> in <code>ncomms_main.tex</code>

6. Tests

Item	Response
Test framework	<code>pytest</code>
Test directory	<code>tests/</code>
Test file inventory	<code>test_core.py</code> (protocol, hashing, provenance, metrics — original 51 tests); <code>test_cost_estimator.py</code> , <code>test_humaneval_loader.py</code> , <code>test_gsm8k_loader.py</code> , <code>test_pass_at_1.py</code> (new for revision)
Total tests	102 (51 original + 51 new revision tests)
Status	All 102 collected; full suite passes locally on Python 3.14.3
How to run	<code>python -m pytest tests/ -v</code>
Coverage targets	Run Card / Prompt Card schema validation, SHA-256 hashing, EMR / NED / ROUGE-L / BERTScore, W3C PROV serialization, HumanEval pass@1 sandbox (timeout, runtime errors, code-fence stripping), GSM8K answer extraction, cost estimator

7. Reproducibility instructions

Step-by-step protocol to reproduce all results in the revised manuscript:

1. Clone and check out the tagged release:

```
git clone https://github.com/Roverlucas/genai-reproducibility-protocol.git
cd genai-reproducibility-protocol
git checkout v1.1
```

2. Create the environment and install pinned dependencies:

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

3. Pull local model weights (open-weight stacks):

```
ollama pull llama3:8b mistral:7b gemma2:9b
```

4. Set API keys for closed stacks (only if reproducing API-served runs):

```
export OPENAI_API_KEY=...
export ANTHROPIC_API_KEY=...
export GEMINI_API_KEY=...
export DEEPSEEK_API_KEY=...
export TOGETHER_API_KEY=...    # quasi-isolation probe
export PERPLEXITY_API_KEY=...  # not used in revision (D4)
```

5. Reproduce the original 4,104 runs:

```
python run_experiments.py          # local stacks + GPT-4 single-turn
python run_claude_multiturn.py     # Claude multi-turn / RAG
python run_gemini_multiturn.py     # Gemini multi-turn / RAG
python run_chat_control.py         # Supplementary SS7 chat-format control
```

6. Reproduce the revision experiments (T1 HumanEval, T1 GSM8K, T4 multi-turn extension, T14 PubMed PM2.5):

```
./run_revision_full.sh             # resumable orchestrator (logs in outputs/revision/logs/)
```

The script honours `--resume` and a per-stack budget cap; partial runs reload from `outputs/revision/checkpoint.json`.

7. Regenerate metrics, tables, and figures:

```
python run_full_analysis.py
python analysis/regenerate_figures_nature_mi.py
python analysis/bertscore_per_field.py      # T6 per-field analysis
python analyze_revision_results.py          # T1+T4+T14 aggregation
```

8. Compile the manuscript:

```
cd article
pdflatex ncomms_main.tex
biber    ncomms_main
pdflatex ncomms_main.tex
pdflatex ncomms_main.tex
```

9. Verify the test suite:

```
python -m pytest tests/ -v          # expect 102 passed
```

Run-level reproducibility evidence is preserved per run in `outputs/runs/<run_id>.json` and `outputs/prov/<run_id>.prov.json` (Run Card + W3C PROV graph; SHA-256 over canonicalised Prompt Card and Run Card).

8. Computing environment

Item	Response
Reference workstation	Apple M4, 24 GB unified memory, macOS 14.6
Python	3.14.3 (CPython); minimum supported 3.10
LLM runtime (local stacks)	Ollama v0.15.5 with Apple Metal acceleration
Container/image	Not required; <code>requirements.txt</code> is the canonical environment specification. A Dockerfile is on the post-acceptance roadmap (Maintenance plan, §10)
OS dependencies	<code>pdflatex</code> , <code>biber</code> (TeX Live or TinyTeX) for manuscript compilation only
Network access	Required only for API-served stacks (OpenAI, Anthropic, Google, DeepSeek, Perplexity, Together AI). Local stacks run fully offline once weights are pulled

Item	Response
Random-seed handling	Per-run seeds (42, 123, 456, 789, 1024) recorded in every Run Card; greedy decoding (temperature 0) used for the deterministic conditions

9. Hardware requirements

Item	Recommended
RAM	16 GB minimum; 24 GB recommended for Gemma 2 9B local serving
Disk	~10 GB (model weights) + ~2 GB (run records, PROV graphs, figures)
GPU	Optional. Apple Silicon (Metal) used for local stacks on the reference workstation; an NVIDIA GPU with 8 GB VRAM achieves equivalent or better throughput. CPU-only execution is supported by Ollama at reduced throughput
API access	Only required to reproduce API-served stacks; ~US\$50 of credit was sufficient for the full revision experimental matrix

10. Maintenance plan

Item	Response
Long-term hosting	GitHub (https://github.com/Roverlucas/genai-reproducibility-protocol). Repository made public on acceptance
Persistent archival	Figshare (DOI: 10.6084/m9.figshare.31653373, CC-BY 4.0). The GitHub release tag v1.1 is mirrored on Figshare. The deposit is available privately to reviewers via the share URL in the Cover Letter and will be made public at acceptance
Issue tracking	GitHub Issues. Bug reports and reproduction questions are accepted from the broader community
Versioning	Semantic versioning. Future protocol changes will not overwrite the manuscript tag
Author commitment	Lucas Rover (corresponding author) commits to maintain the repository for 5 years post-publication, in accordance with the funding institution’s data-management policy (UTFPR)
Backup	Mirror to institutional repository (UTFPR / RIUT) on acceptance

11. Items requiring final insertion before submission

Item	Status	Action
Figshare DOI		10.6084/m9.figshare.31653373 — already minted (CC-BY 4.0); private reviewer share URL: https://figshare.com/s/3d17327ceflae99ed37c
Final run count (4,104 original + 2,900 revision = 7,004)		Reported in manuscript Table 1, Methods §Tasks, and Table 4 (revision tasks)

Item	Status	Action
Pass@1 results (HumanEval)		Reported in Methods §“Coding and math reasoning” and Table 4
Final tagged release v1.1		Created on GitHub at commit <code>d60ca2d</code> (origin/main); referenced in Data Availability and Code Availability statements
Reviewer figshare link	Provided to editorial office; not in the public manuscript	https://figshare.com/s/3d17327cef1ae99ed37c (private; for reviewers only)

Prepared by Lucas Rover (corresponding author). Cross-references: `REVISION_PLAN.md`, `STATUS.md`, `article/REPORTING_SUMMARY.md`, `article/ML_CHECKLIST_FILLED.md`.